

cover_tree_tableau_ALCIRCC5.tex
completeness_extraction_ALCIRCC5.tex
alcircc5_cover_tree_tableau_needall_patched.tex
sibling_interface_descriptors_ALCIRCC5_completed_eqsync_canonical_needpatched.tex

Thanks, these are the right files. I'll check the theorem chain across the TeX proofs: split-forest soundness, finite quotient extraction, EQ collapse, and CT/tableau equivalence. I've found a major issue in the completeness extraction: it assumes well-founded Hasse-style PP covers, which arbitrary RCC5 models need not provide. I reviewed the added TeX files. My revised verdict is stronger than before:

The new files fill in some intended details, but they introduce or expose several proof-level gaps. The largest one is not just a missing lemma; it is a structural problem with the rooted cover-tree/split-tree assumption.

Most serious obstruction: rooted cover trees lose satisfiable upward-PP models

That assumption is not valid for formulas that force an infinite upward PP chain.

$$C_{\uparrow} := \exists \backslash PP. \top \sqcap \forall \backslash PP. \exists \backslash PP. \top.$$

1. the evaluation point has a strict PP-ancestor/superpart; and
2. every strict PP-ancestor/superpart has a further strict PP-ancestor/superpart.

$$a_0 \setminus \text{PP } a_1 \setminus \text{PP } a_2 \setminus \text{PP } a_3 \setminus \text{PP } \cdots$$

But no rooted cover tree of the kind defined in the proof can represent such a model. If the evaluation node is below an ambient root ϱ , then the evaluation node is PP-related to ϱ . By $\forall \backslash \text{PP}. \exists \text{PP}. \top$, the root ϱ must itself have a strict PP-ancestor. But by definition the ambient root has none.

So the completeness direction fails unless the proof is changed to allow rootless, upward-infinite, or bi-infinite regular split structures, or unless a nontrivial conservativity theorem shows how to add an ambient root without breaking PP-universals. The current files do not provide that.

1. The PP-DAG/Hasse construction is formally invalid

In completeness_extraction...tex, lines 283-287, the proof says the strict PP order is a “well-founded DAG.” A strict partial order need not be well-founded. It may contain infinite ascending or descending chains. The proof then unfolds the “immediate-PPI relation,” i.e. the Hasse diagram, at

lines 297-299. But arbitrary partial orders need not have immediate covers. Dense proper-part orders have no Hasse edges at all.

There is also an orientation problem. The text defines $d_1 \leq_{\text{PP}} d_2$ iff $d_1 \setminus \text{PP } d_2$, then chooses a “minimal” element as ambient root, and even proposes adding $d_\top$ with $\rho(d_\top, d) = \text{PP}$ for all d , lines 289-295. Under the paper’s own orientation, that makes $d_\top$ a universal proper part, not a top/superpart. A cover-tree root should be a maximal superpart, i.e. PPI-related downward to descendants.

This is not a cosmetic issue; it affects the existence of the split-tree presentation itself.

2. The descriptor-extraction proof conflates different witnesses

The completeness extraction defines the relation table M_k by aggregating all actually realized relations between endpoint summaries; see completeness_extraction...tex, lines 377-386.

Then, in the proof of C3 downward support, lines 469-485, it argues roughly:

if summary p can evolve to p' , and relation R appears between summaries p and q , then compose the child of x with y to get support for R .

But the x that has a child of summary p' need not be the same occurrence of summary p that realizes relation R to q . Since M_k aggregates by state/status, R may be witnessed by one occurrence of p , while the $p \rightarrow p'$ child-step is witnessed by another occurrence of p . The proof silently identifies them.

The same problem appears in C5 triple coherence, lines 491-505. The proof takes $R_{12} \in M(\chi_1, \chi_2)$ and $R_{13} \in M(\chi_1, \chi_3)$, then assumes there is one triple x_1, x_2, x_3 realizing both relations. But R_{12} and R_{13} may be realized by different occurrences of the same summary χ_1 . Aggregation by finite state does not preserve this kind of joint witness information automatically.

This is a standard amalgamation/support-closure problem. The earlier split-forest note says the descriptor tables should be “least locally coherent support-closed tables” containing realized relations (sibling_interface...tex, line 690), but the completeness extraction does **not** actually construct such a closure. It simply uses realized atomic relations. That is insufficient.

3. PP/PPI eventualities are handled inconsistently

The split-forest state definition says the witness menu records “for each existential one designated target color and target relation domain” (sibling_interface...tex, line 367). The patched tableau note also says the witness menu handles ReqUp and ReqDown, i.e. PP/PPI eventualities (alcirc5_cover_tree_tableau_needall_patched...tex, line 297).

But the completeness extraction says the opposite: if $R \in \{\setminus \text{PP}, \setminus \text{PPI}\}$, the witness is handled by the tree structure and “does not enter the witness menu” (completeness_extraction...tex, lines 399-401). That inconsistency matters. For transitive roles, a witness for $\exists \setminus \text{PP}. C$ or $\exists \setminus \text{PPI}. C$ may be arbitrarily far away. Parent/child slots alone do not guarantee eventual realization after quotienting and blocking. A finite quotient needs an explicit eventuality condition, progress/ranking condition, or Büchi-style acceptance argument. The files do not give one.

This is closely related to the $C_\top$ example above.

4. Strong-EQ/core matching remains under-specified

The split proof stores core multiplicities only as

$$K_k: \Sigma_k \rightarrow \{0, 1, +\}$$

in sibling_interface...tex, lines 299-302. Later, the unfolding says core matches are unfolded as distinct EQ-mates (sibling_interface...tex, line 431), and the network construction claims that designated-mate recursion synchronizes all domains to third nodes, lines 447-459.

But the finite data shown do not specify a globally consistent matching of core copies across all sibling interfaces. Same rank- d state does not mean same semantic element. Strong EQ quotienting requires an actual equivalence relation that is a typed congruence; the proof defines typed congruence at lines 611-619 and proves quotienting under that assumption at lines 671-677, but the finite certificate does not yet show how the congruence is globally chosen and checked.

5. The finite validity check is not fully formalized

The decidability theorem depends on the claim that valid finite quotients are effectively enumerable and checkable. The proof sketch says this at `sibling_interface...tex`, lines 681-690. But the validity conditions are still partly schematic:

- C5 “triple coherence” is described informally in `sibling_interface...tex`, lines 342-343.
- “Every induced pair-domain created from witness-menu edges by composition with tree edges” is required in `sibling_interface...tex`, lines 402-407, but the finite closure procedure is not precisely defined.
- The operational side-checker explicitly includes “every finite prefix network built from the quotient has a canonical extendible refinement” as a condition (`alcircc5_cover_tree_tableau_needall_patched...tex`, lines 339-346). That is infinitary unless reduced to a finite condition.

The soundness direction has a plausible finite-prefix/König argument, but the finite certificate language still needs a precise decidable definition.

6. The implementation paper itself still admits the algorithmic equivalence gap

The implementation report is candid that CT1-CT5 are not formally derived from the split-forest quotient semantics. It says the tableau is “not a formal derivation” of the semantics and that CT5 was added by empirical counterexample analysis; see `cover_tree_tableau_ALCIRCC5(1).tex`, lines 964-1008. It also lists formal correctness as an open question at lines 1013-1021.

So even if the abstract quotient theorem were fixed, the CT1-CT5 implementation would still not yet be a proved decision procedure.

What looks useful or salvageable

There are real useful pieces here:

- The separation of tree PP/PPI structure from sibling DR/PO cross-edges is a good idea.
- The observation that open non-core sibling cross-edges reduce to {DR, PO} after splitting is plausible.
- The typed-EQ quotient idea is the right kind of condition for restoring strong equality.
- The relation-sensitive Need_R filtering is necessary; the patched files correctly identify the previous unsoundness around composition-induced PP/PPI labels.
- The finite-prefix/König construction may be salvageable once the finite certificate is made precise.

But the current completeness proof does not establish that every satisfiable concept has such a finite rooted quotient.

Bottom line

I would not accept the current decidability theorem.

The main theorem would need at least the following repairs:

1. Replace the rooted Hasse cover-tree assumption with a structure that can represent infinite upward PP chains, or prove a conservative-root-extension theorem.
2. Remove the false “strict PP order is well-founded” step and avoid relying on immediate covers unless a cover relation is proven to exist.
3. Give a genuine support-closed descriptor extraction, not just aggregation of realized relations.
4. Add a formal eventuality mechanism for transitive PP/PPI witnesses after quotienting.
5. Specify a finite, decidable validity condition for quotients, including C3-C5, induced-domain closure, and EQ-core matching.
6. Separately prove that CT1-CT5 is equivalent to the finite-quotient validity notion, or present CT1-CT5 only as empirical implementation evidence.

The most compact counterexample to address first is:

$$\exists \backslash PP. \top \sqcap \forall \backslash PP. \exists \backslash PP. \top .$$

Any corrected completeness proof should explain exactly how its finite certificate represents this satisfiable rootless-upward behavior.